



# Functional Description

Electronics

Interface Specification SAE J1939

ECU 8 – FSW Vers. 1.3 ... 1.4

ECU 9 Diesel – FSW Vers. 1.5 ... 2.8

ECU 9 Gas – FSW Vers. 2.3.X ... 2.7.X

ECU 9 BiFuel – FSW Vers. 1.0

plus Engine Monitoring Unit EMU 8

and ECU Extension Unit EXU 1

E532424/06E



A Rolls-Royce  
solution

© 2019 Copyright MTU Friedrichshafen GmbH

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of MTU Friedrichshafen GmbH. This particularly applies to its reproduction, distribution, editing, translation, micro-filming and storage or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. MTU Friedrichshafen GmbH reserves the right to change, delete or supplement the information provided as and when required.

# Table of Contents

|        |                                                                               |     |
|--------|-------------------------------------------------------------------------------|-----|
| 1      | General Information                                                           |     |
| 1.1    | General                                                                       | 6   |
| 1.2    | References                                                                    | 7   |
| 1.3    | SAE J1939 introduction and network architecture                               | 8   |
| 1.4    | J1939 Message Terminology                                                     | 10  |
| 1.5    | About the structure of this document                                          | 11  |
| 1.6    | Release notes                                                                 | 12  |
| 2      | Interface Description                                                         |     |
| 2.1    | Supported “Parameter Group Numbers” - (PGN)                                   | 14  |
| 3      | PGNs of Engine Control Unit ECU                                               |     |
| 3.1    | General Parameter Groups                                                      | 18  |
| 3.1.1  | PGN 59904 - Request                                                           | 18  |
| 3.1.2  | PGN 59302 - ACK/NACK - Positive / Negative (Acknowledgment)                   | 19  |
| 3.1.3  | PGN 60416 - TP.CM_BAM - Broadcast Announce                                    | 21  |
| 3.1.4  | PGN 60160 - TP.DT - Data packet (Data Transfer)                               | 22  |
| 3.1.5  | PGN 60928 - ACL - (Address Claimed / Cannot Claim)                            | 23  |
| 3.2    | Parameter Group Numbers PGN                                                   | 24  |
| 3.2.1  | PGN 0 - Torque/Speed Control 1 (TSC1)                                         | 24  |
| 3.2.2  | PGN 52992 - Continuous Torque & Speed Limit Request (CTL)                     | 29  |
| 3.2.3  | PGN 56832 - Reset (RESET)                                                     | 30  |
| 3.2.4  | PGN 57344 - Cab Message 1 (CM1)                                               | 32  |
| 3.2.5  | PGN 61440 - Electronic Retarder Controller (ERC1)                             | 36  |
| 3.2.6  | PGN 61441 - Electronic Brake Controller 1 (EBC1)                              | 38  |
| 3.2.7  | PGN 61442 - Electronic Transmission Controller 1 (ETC1)                       | 42  |
| 3.2.8  | PGN 61443 - Electronic Engine Controller 2 (EEC2)                             | 44  |
| 3.2.9  | PGN 61444 - Electronic Engine Controller 1 (EEC1)                             | 46  |
| 3.2.10 | PGN 61445 - Electronic Transmission Controller 2 (ETC2)                       | 51  |
| 3.2.11 | PGN 61454 - Aftertreatment 1 Intake Gas 1 (AT1IG1)                            | 52  |
| 3.2.12 | PGN 61455 - Aftertreatment 1 Outlet Gas 1 (AT1OG1)                            | 54  |
| 3.2.13 | PGN 61456 - Aftertreatment 2 Intake Gas 1 (AT2IG1)                            | 56  |
| 3.2.14 | PGN 61457 - Aftertreatment 2 Outlet Gas 1 (AT2OG1)                            | 58  |
| 3.2.15 | PGN 61475 - Aftertreatment 1 SCR Dosing System Information 1 (A1SCRDSI1)      | 60  |
| 3.2.16 | PGN 61476 - Aftertreatment 1 SCR Dosing System Requests 1 (A1SCRDSR1)         | 62  |
| 3.2.17 | PGN 61478 - Aftertreatment 2 SCR Dosing System Information 1 (A2SCRDSI1)      | 64  |
| 3.2.18 | PGN 61479 - Aftertreatment 2 SCR Dosing System Requests 1 (A2SCRDSR1)         | 66  |
| 3.2.19 | PGN 64660 - Aftertreatment System Abuse Information (ASAI)                    | 68  |
| 3.2.20 | PGN 64671 - Dash Display 2 (DD2)                                              | 69  |
| 3.2.21 | PGN 64672 - Multi Fuel Engine Hours (MFEH)                                    | 70  |
| 3.2.22 | PGN 64701 - Aftertreatment Trip Diesel Exhaust Fluid (A1SCRDSI2)              | 71  |
| 3.2.23 | PGN 64712 - Electronic Engine Controller 13 (EEC13)                           | 72  |
| 3.2.24 | PGN 64824 - Aftertreatment 2 SCR Exhaust Gas Temperature 1 (A2SCREGT1)        | 74  |
| 3.2.25 | PGN 64827 - Aftertreatment 2 SCR Dosing System Information 2 (A2SCRDSI2)      | 75  |
| 3.2.26 | PGN 64829 - Aftertreatment 1 SCR Reagent Tank 2 Information (A1SCRRT2I)       | 78  |
| 3.2.27 | PGN 64830 - Aftertreatment 1 SCR Exhaust Gas Temperature 1 (A1SCREGT1)        | 80  |
| 3.2.28 | PGN 64833 - Aftertreatment 1 SCR Dosing System Information 2 (A1SCRDSI2)      | 81  |
| 3.2.29 | PGN 64851 - Engine Average Information (EAI)                                  | 84  |
| 3.2.30 | PGN 64870 - Engine Temperature 4 (ET4)                                        | 85  |
| 3.2.31 | PGN 64912 - Advertised Engine Torque Curve (AETC)                             | 86  |
| 3.2.32 | PGN 64914 - Engine Operating Information (EOI)                                | 87  |
| 3.2.33 | PGN 64915 - Generator Control 1 (GC1)                                         | 91  |
| 3.2.34 | PGN 64929 - Aftertreatment 1 Fuel Control 1 (AT1FC1)                          | 93  |
| 3.2.35 | PGN 64930 - Fuel information 3 Gaseous (GFI3)                                 | 95  |
| 3.2.36 | PGN 64931 - Electronic Engine Controller 6 (EEC6)                             | 96  |
| 3.2.37 | PGN 64938 - Engine Fluid Level/Pressure 4 (MEC7)                              | 98  |
| 3.2.38 | PGN 64965 - ECU Identification Information (ECUID)                            | 99  |
| 3.2.39 | PGN 64971 - Off-Highway Engine Control Selection (OHECS)                      | 100 |
| 3.2.40 | PGN 64976 - Inlet/Exhaust Conditions 2 (IC2)                                  | 103 |
| 3.2.41 | PGN 64992 - Ambient Conditions 2 (AMB2)                                       | 104 |
| 3.2.42 | PGN 65099 - Transmission Configuration 2 (TCFG2)                              | 105 |
| 3.2.43 | PGN 65110 - Aftertreatment 1 Diesel Exhaust Fluid Tank 1 Information (AT1T1I) | 106 |
| 3.2.44 | PGN 65128 - Vehicle Fluids (VF)                                               | 109 |
| 3.2.45 | PGN 65130 - Engine Fuel/lube systems (EFS)                                    | 110 |
| 3.2.46 | PGN 65133 - Heater Information (HTR)                                          | 113 |
| 3.2.47 | PGN 65153 - Fuel Information 2 (Gaseous)                                      | 115 |



|      |                                                     |     |      |                                                                                        |     |
|------|-----------------------------------------------------|-----|------|----------------------------------------------------------------------------------------|-----|
| 4.6  | PGN 65183 - Exhaust Port Temperature 5 (EPT5)       | 223 | 4.19 | PGN 65411 - MTU Exhaust Temperature Limits (METL)                                      | 236 |
| 4.7  | PGN 65184 - Exhaust Port Temperature 4 (EPT4)       | 224 | 5    | Settings                                                                               |     |
| 4.8  | PGN 65185 - Exhaust Port Temperature 3 (EPT3)       | 225 | 5.1  | CAN source addresses                                                                   | 237 |
| 4.9  | PGN 65186 - Exhaust Port Temperature 2 (EPT2)       | 226 | 5.2  | J1939 parameters of the ECU and ECU 9                                                  | 238 |
| 4.10 | PGN 65187 - Exhaust Port Temperature 1 (EPT1)       | 227 | 6    | Fault List                                                                             |     |
| 4.11 | PGN 65188 - Engine Temperature 2 (ET2)              | 228 | 6.1  | General information about available fault codes (DTC)                                  | 239 |
| 4.12 | PGN 65246 - Air Start Pressure (AIR2)               | 229 | 6.2  | List of available ECU fault codes (Part 1)                                             | 240 |
| 4.13 | PGN 65271 - Vehicle Electrical Power (VEP)          | 230 | 6.3  | List of available ECU fault codes (Part 2)                                             | 252 |
| 4.14 | PGN 65363 - MTU Eng Fluid Level Pressure 1 (MEFLP1) | 231 | 6.4  | List of available ECU fault codes (Part 3)                                             | 265 |
| 4.15 | PGN 65364 - MTU Engine Temperature 1 (MET1)         | 232 | 6.5  | List of available ECU fault codes (Part 4)                                             | 279 |
| 4.16 | PGN 65369 - MTU Splash Oil Temperature 5 (MSOT5)    | 233 | 6.6  | List of available EXU fault codes (individual accumulator pressure measuring function) | 286 |
| 4.17 | PGN 65370 - MTU Splash Oil Temperature 4 (MSOT4)    | 234 | 7    | Appendix A                                                                             |     |
| 4.18 | PGN 65371 - MTU Splash Oil Temperature 3 (MSOT3)    | 235 | 7.1  | Abbreviations                                                                          | 291 |
|      |                                                     |     | 7.2  | MTU Contact/Service Partners                                                           | 292 |
|      |                                                     |     | 7.3  | Conversion tables                                                                      | 293 |

# 1 General Information

## 1.1 General

### Definition of terms

This document defines and describes the SAE J1939 interface of the ECU for the following applications:

- Rail
- C&I
- Genset
- Oil&Gas

The term “interface” in this context is limited to the data or information interface of the ECU on the SAE J1939 CAN bus. Not included in this definition is the interface specification of the ECU via PCS-5 CAN bus and the exchange of SAE J1939 data and information between other SAE J1939 network components which are not involved in the ECU SAE J1939 interface (i.e. other devices connected to the CAN bus which neither communicate with nor “listen-in” to the ECU).

### Document structure and principles

This interface specification is divided into the following sections for reasons of intelligibility and clarity:

- Supported PGNs
- General parameters
- PGNs in detail
- Settings at the ECU
- DM1 and DM2 (fault/error list)

The interface definitions in this document are based on the producer-consumer principle. Hence, a component in the SAE J1939 network produces (and transmits) data or information while simultaneously consuming (receiving) data and information from other components (producers). A component can thus be seen as acting both as a producer of certain data as well as a consumer of other data.

Based on this principle, so-called communication matrices have been established for CAN parameter groups and signals. The abbreviations T for Transmitter (= producer) and R for Receiver (= consumer) are used. This information is stated in the first columns explaining the PGNs.

The following principles apply to the representation of numbers in this document:

- All numbers are represented in decimal notation (e.g. physical variables) unless explicitly stated otherwise.
- The hexadecimal system is used to represent data bytes (CAN message payloads) or CAN identifiers, etc. In such cases, the numerical value is followed immediately by an “h” identifying it as hexadecimal.
- Numerical values such as bit assignments may also be represented using the binary system in addition to the hexadecimal system. Such numbers are identified by the suffix “b”.

#### Important!

For engines subject to US EPA 40 CFR 1039 / CARB emissions emission standards, the J1939 message (up to and including ECU 9 Diesel SW V. 2.3) PGN 65226 DM1, SPN 521506, FMI 31, must be displayed to the operator with active AECD "Engine Overheat Protection EGR Cooler Overheat Protection". As of ECU 9 Diesel SW V. 2.4 and higher, there is the alternative option of displaying the status PGN 65444 SPN 521808 AECD active. The display of the J1939 message can only be omitted if the activation of the AECD message with a wired signal lamp is displayed (for further details see MTU scheme wiring).

TIM-ID: 0000049106 - 002

## 1.2 References

All documents listed here may be implemented in full, in part or with changes in content or structure.

### Applicable standard documents

The following standard documents shall serve as basic standards in addition to this document. This document shall take precedence over the documents listed below in case of conflict. The latest revision shall be used if no specific version of the following standard documents is stated.

| Designator     | Standard document                                                                      | Revision        |
|----------------|----------------------------------------------------------------------------------------|-----------------|
| [SAE J1939]    | SAE J1939 Recommended Practice for a Serial Control and Communications Vehicle Network | Revised 2013-08 |
| [SAE J1939-21] | SAE J1939 Data Link Layer                                                              | Revised 2010-12 |
| [SAE J1939-71] | SAE J1939 Vehicle Application Layer                                                    | Revised 2013-09 |
| [SAE J1939-73] | SAE J1939 Application Layer – Diagnostics                                              | Revised 2013-07 |
| [SAE J1939-75] | SAE J1939 Application Layer – Generator Sets and Industrial                            | Revised 2011-05 |
| [SAE J1939-81] | SAE J1939 Network Management                                                           | Revised 2011-06 |

SAE publications are available from:

- SAE  
400 Commonwealth Drive  
Warrendale, PA 15096-0001  
Telephone: 877-606-7323 (in the USA and Canada) or 724-776-4970 (outside the USA)

More information is available at:

- [www.sae.org](http://www.sae.org)

## 1.3 SAE J1939 introduction and network architecture

A CAN bus (CAN 2) with a baud rate of 250 Kbit/s is used for data interchange between the ECU and other components. Data is transmitted in accordance with the SAE J1939 standard (see [SAE J1939]).

The MTU dialog system DiaSys has the task of:

- Acquiring internal measured values
- Calibrating, reading out and archiving internal fault memories
- Uploading and downloading ECU software and parameters.

DiaSys communicates with the ECU on CAN 1 with the proprietary dialog protocol FB1 DP.

The proprietary MTU CAN protocol PCS-5 is also used on the ECU CAN 1. Parallel or redundant utilization of both CAN busses is not envisaged. The system integrator must decide on a communication procedure which is then deposited in the engine parameter record when the ECU is delivered. This can be switched at a later date by setting the parameters of the ECU accordingly using the dialog system DiaSys. Changes only take effect after restarting the ECU. CAN 1 is used for communication with DiaSys (dialog protocol FB1 DP) even when CAN 2 is utilized with SAE J1939.

SAE J1939 defines a communication network based on a highspeed CAN bus. The SAE J1939 standard describes layers 2 to 7 of the ISO/OSI layer model (see [SAE J1939]). The CAN messages in the SAE J1939 network use the extended message format (Extended Frame Format, see [SAE J1939-21] for details).

The term parameter group (see [SAE J1939] section 2.2.1) is used to define an atomic data packet. The term parameter group is used in SAE J1939:

- To designate simple CAN messages
- For segmented CAN messages with more than 8 bytes of payload data (so-called “multipacket” messages).

See [SAE J1939-21] section 5.10 for details.

SAE J1939 uses node addresses (Source Address, SA) to identify and differentiate components (control units, controllers) in the network. A node address in an SAE J1939 network must be unique. These node addresses are used in the CAN identifier to determine which component is the transmitter (source address) of a parameter group. In conjunction with the parameter group number (PGN), the source address forms the CAN identifier (details are provided in [SAE J1939-21]).

The ECU uses the Service Configurable Address CA claiming method (see [SAE J1939-81], section 3.3.1.2). Both the ECU source address and the source address of the generator controller can be configured in ECU using the diagnostics system DiaSys. The default source address of the ECU is 00h, the default source address of the generator controller is defined as EAh. These default values conform with [SAE J1939], Table B2 and Table B7. These changes only take effect after restarting the ECU (see section “ACL – (Address Claimed / Cannot Claim)”).

Some signals defined in the SAE J1939 standard (particularly SAE J1939-71) are identified using the term “Suspect Parameter Number” (SPN). The term “suspect parameter” is used synonymously with the term “signal” in this document.

Various signals exchanged between the ECU and other components are not defined in the SAE J1939-71 standard (see [SAE J1939-71]). Proprietary parameter groups are used to transmit these signals in the context of the SAE J1939 protocol (Proprietary B Parameter Group, see [SAE J1939-21] sections 5.4.5). Some of these parameter groups require more than an 8 byte payload and therefore resort to the SAE J1939 transport protocol (TP, segmented messages, see [SAE J1939-21] section 5.10).

All SAE J1939 standard parameter groups used in this document are described in accordance with SAE J1939-71 Standards (see [SAE J1939-71]). They may include signals (parameters) which are not used by the ECU. These unused signals must be transmitted as “Not Available”, see also [SAE J1939-71] section 5.4.1). The communication relationship of such signals is not stated (no T and no R) in the relevant communication matrix of this document. The signal neither transmitted nor received by any component and is thus considered “Not Available”. Undefined or reserved signals (payload) in a parameter group are designated “Reserved”. These also have no communication relationship. Any signal having no communication relationship is to be considered “Not Available” (see “Data formats” below for details concerning “Not Available”).



The following applies: A component generally does not receive messages it has transmitted itself. The component does not receive the messages it has transmitted itself if both transmission and receiving are defined for the signals of a message for a component, . It receives the message transmitted by another component.

Two transport protocols are defined in the SAE J1939 standard. The “broadcast transport protocol” (BAM transport protocol) is used for the non-directional transmission of segmented messages (refer to [SAE J1939-21], section 5.10.2.1 for details of “broadcasting”). The directional transmission transport protocol (RTS/CTS transport protocol) is used for directional transmission of segmented messages. In this case, a virtual connection is established between two nodes, the transmitter and the receiver (refer to [SAE J1939-21], section 5.10.2 for details).

The BAM transport protocol is used for the transmission of segmented messages to-and-from the ECU.

## Data formats

SAE J1939 defines:

- The value range of a valid signal
- Error indication
- The ID range of signals which are “not available”.

The actual characteristics of these value ranges depend on the data length of the signals and are listed in the table below. Refer to [SAE J1939-71], section 5.1.5 for details.

| Range         | 2 bits    | 1 byte    | 2 bytes       | 4 byte                | ASCII     |
|---------------|-----------|-----------|---------------|-----------------------|-----------|
| Valid signal  | 00h – 01h | 00h – Fah | 00h – FAFh    | 00000000h – FFFFFFFFh | 01h – Feh |
| Error         | 02h       | Feh       | FE00h – FEFFh | FE000000h – FFFFFFFFh | 00h       |
| Not available | 03h       | FFh       | FF00h – FFFFh | FF000000h – FFFFFFFFh | FFh       |

Signals with a data length  $\geq 2$  bytes are transmitted in Little Endian (Intel, byte with the lowest-order bit first) byte order (see [SAE J1939-71], section 5.1.2).

The 32-bit integer 439.041.101 (1A2B3C4Dh) is thus transmitted as follows:

| Byte | Value                    |
|------|--------------------------|
| 1    | 4Dh Low Byte, Low Word   |
| 2    | 3Ch High Byte, Low Word  |
| 3    | 2Bh Low Byte, High Word  |
| 4    | 1Ah High Byte, High Word |

## Network management (start procedure)

The ECU uses the Service Configurable Address CA claiming method (see [SAE J1939-81], section 3.3.1.2). The ECU thus meets SAE J1939 requirements for network management (see [SAE J1939-81], section 4.5). This means that ECU source address assignment is static. The ECU assignment cannot be dynamically changed by “Address Claimed / Cannot Claim” (cf. section “ACL – (Address Claimed / Cannot Claim)”). Both the ECU source address and the source address of the generator controller can be configured in the ECU using the diagnostics system DiaSys (see section “Source addresses”, (→ Page 237)). These changes only take effect after restarting. Like every other component in the J1939 network, the ECU has a unique network-wide identifier, the NAME (see section “ACL – (Address Claimed / Cannot Claim)”).

## 1.4 J1939 Message Terminology

### Parameter

A specific data element within a PGN message.

Each parameter has a description, resolution, range, scaling, offset and data size defined by the vendor. Standard parameters are defined in the SAE J1939-71 specification.

### PGN

Parameter Group Number

Parameters are grouped by common purpose and assigned a PGN. The PGN is used in J1939 messages to identify the group, and hence the parameter data, that is contained in the message.

### SPN

Suspect Parameter Number

Each parameter is assigned an SPN. The SPN is used in diagnostics to specify the particular item that the service code is associated with.

### PGN Transmission Rate

The nominal rate at which the PGN message will be transmitted.

The PGN definition specifies the transmission rate of the PGN message. Rates specified as “On Request” are only transmitted when a request for a PGN is received from another network node.

### Parameter Offset

The offset into the PGN message buffer where a parameter’s data is located.

A PGN message contains the data for all of the parameters in its associated parameter group. The PGN definition indicates where each parameter’s data is located, based on offsets from the front of the message buffer. Note that offsets may not be on byte boundaries but may be bit based.

### Parameter Length

The length of an individual parameter’s data.

A parameter’s data may be contained in a single bit to multiple bytes within the PGN message buffer. Note that data length is not always in bytes but may be a number of bits.

### Parameter Resolution

The scale and offset of a parameter’s data.

The parameter data included in a PGN message is raw unsigned integer data. The scale and offset define the translation to convert the raw data to engineering units.